GREEEN tallgrass prairie restoration: RLQ analyses pool size (SW Station)

MB

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Preparation

```
library(here)
library(tidyverse)
library(ade4)
```

Packages

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types =
    cols(
    .default = "?",
    seeding_time = "f"
  )) %>%
  arrange(id_plot_year) %>%
  filter(
    site == "SW Station",
```

```r
    richness_type == "seeded_richness",
    !(treatment_id %in% c("2", "4"))
  ) %>%
  mutate(
    treatment = str_c(seeding_time, seeded_pool, sep = "_"),
    treatment = factor(treatment)
  ) %>%
  select(
    id_plot_year, year, treatment, water_cap, cover_non_seeded,
    cover_seeded_grass
    )

traits <- read_csv(
  here("data", "processed", "data_processed_traits.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(.default = "?")
  ) %>%
  filter(seeded == "1", family != "Poaceae") %>%
  select(accepted_name, sla, height, seedmass) %>%
  mutate(seedmass = log(seedmass)) %>%
  drop_na()

species <- read_csv(
  here("data", "processed", "data_processed_species.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(.default = "?")
  ) %>%
  filter(plot_size %in% c("1", "25")) %>%
  mutate(abundance = 1) %>%
  select(id_plot_year, accepted_name, abundance) %>%
  semi_join(sites, by = "id_plot_year") %>%
  semi_join(traits, by = "accepted_name") %>%
  arrange(id_plot_year)
```

Load data

Checks

```r
sites %>%
  group_by(year) %>%
  count()
```

Plots per year

```
## # A tibble: 4 x 2
## # Groups:   year [4]
##    year     n
##   <dbl> <int>
## 1  2015    48
## 2  2016    48
## 3  2017    48
## 4  2018    48
```

```r
species %>%
  group_by(id_plot_year) %>%
```

```r
  count(name = "seeded_forb_richness") %>%
  ungroup() %>%
  count(seeded_forb_richness, name = "number_of_plots") %>%
  mutate(
    "plots_>=_1_species" = sum(number_of_plots),
    total_number_plots = 72
    )
```

Seeded species richness

```
## # A tibble: 9 x 4
##   seeded_forb_richness number_of_plots `plots_>=_1_species` total_number_plots
##                  <int>           <int>                 <int>              <dbl>
## 1                    1              36                   173                 72
## 2                    2              34                   173                 72
## 3                    3              34                   173                 72
## 4                    4              30                   173                 72
## 5                    5              11                   173                 72
## 6                    6              13                   173                 72
## 7                    7               8                   173                 72
## 8                    8               5                   173                 72
## 9                    9               2                   173                 72
```

```r
traits %>% anti_join(species, by = "accepted_name")
```

Check non-occuring seeded species

```
## # A tibble: 13 x 4
##    accepted_name                   sla height seedmass
##    <chr>                         <dbl>  <dbl>    <dbl>
##  1 Baptisia alba                 268.   0.940    -4.12
##  2 Eryngium yuccifolium           91.7  1.01     -5.57
##  3 Liatris cylindracea           110.   0.564    -5.73
##  4 Penstemon hirsutus            148.   0.457    -8.74
##  5 Silphium terebinthinaceum     171.   0.716    -4.53
##  6 Solidago juncea               130.   0.783    -7.79
##  7 Solidago nemoralis            122.   0.61     -7.84
##  8 Solidago rigida                86.5  1.1      -7.27
##  9 Solidago speciosa             263.   0.71     -8.27
## 10 Symphyotrichum laeve          127.   0.975    -7.76
## 11 Symphyotrichum oolentangiense 143.   0.362    -8.68
## 12 Tradescantia ohiensis         103.   0.921    -5.61
## 13 Zizia aptera                  100.   0.34     -6.42
```

Finalize preparation

```r
traits <- traits %>%
  semi_join(species, by = "accepted_name") %>%
  column_to_rownames(var = "accepted_name")

sites <- sites %>%
  semi_join(species, by = "id_plot_year") %>%
  column_to_rownames(var = "id_plot_year")
```

```r
species <- species %>%
  pivot_wider(
    names_from = "accepted_name", values_from = "abundance", values_fill = 0
    ) %>%
  column_to_rownames(var = "id_plot_year")
```

Statistics

Descriptive analyses

```r
sites %>%
  group_by(year) %>%
  count()
```

```
## # A tibble: 4 x 2
## # Groups:   year [4]
##    year     n
##   <dbl> <int>
## 1  2015    37
## 2  2016    45
## 3  2017    46
## 4  2018    45
```

Ordinations

```r
dudi_l <- dudi.coa(species, scannf = FALSE)
dudi_r <- dudi.hillsmith(
  sites,
  row.w = dudi_l$lw,
  scannf = FALSE
  )
dudi_q <- dudi.pca(
  traits,
  row.w = dudi_l$cw,
  scannf = FALSE
  )
```

RLQ analyses

```r
m_rlq <- rlq(
  dudiR = dudi_r,
  dudiL = dudi_l,
  dudiQ = dudi_q,
  scannf = FALSE
  )
```

Global test

Model 2: species-environment relationship

Model 4: species-traits relationship

```r
randtest(m_rlq, nrepet = 999, modeltype = 6)
```

```
## class: krandtest lightkrandtest
## Monte-Carlo tests
## Call: randtest.rlq(xtest = m_rlq, nrepet = 999, modeltype = 6)
##
## Number of tests:   2
##
## Adjustment method for multiple comparisons:   none
## Permutation number:   999
##      Test       Obs   Std.Obs   Alter Pvalue
## 1 Model 2 0.2755483 22.531026 greater  0.001
## 2 Model 4 0.2755483  0.917171 greater  0.179
```

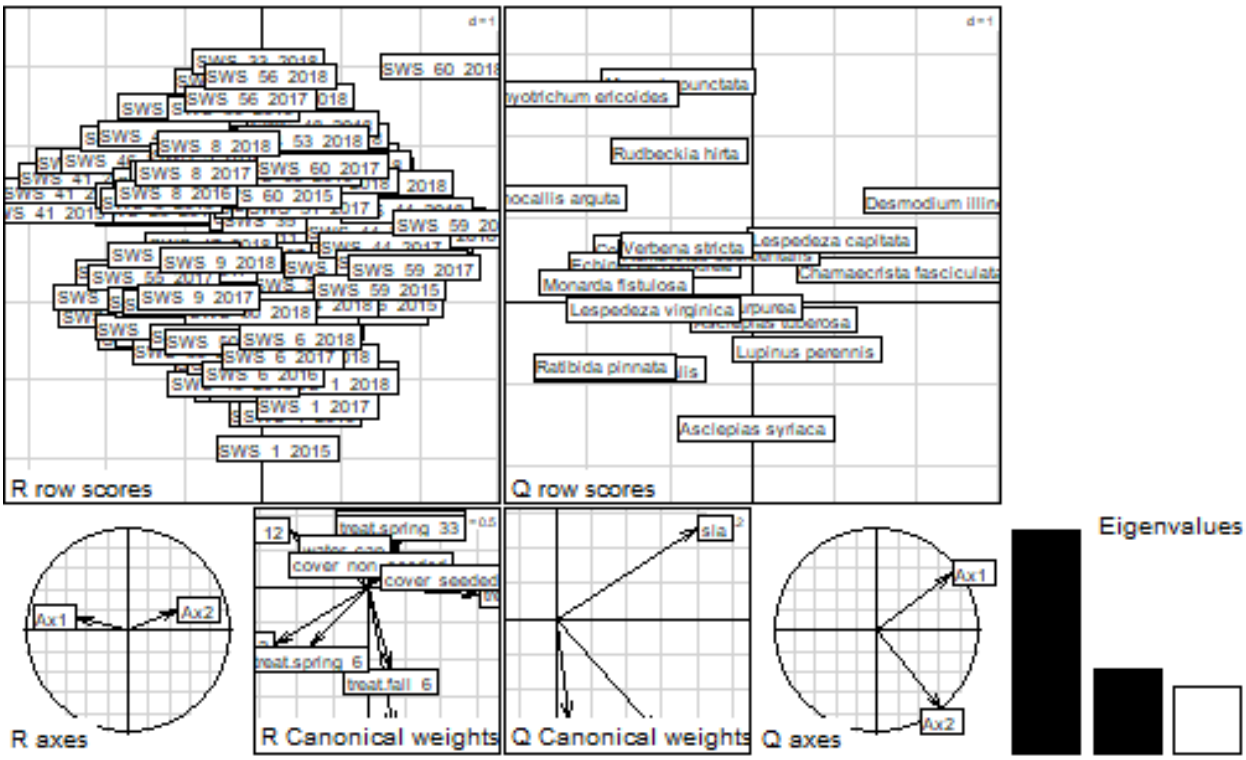
Traits by environment crossed table

```r
m_rlq$tab
```

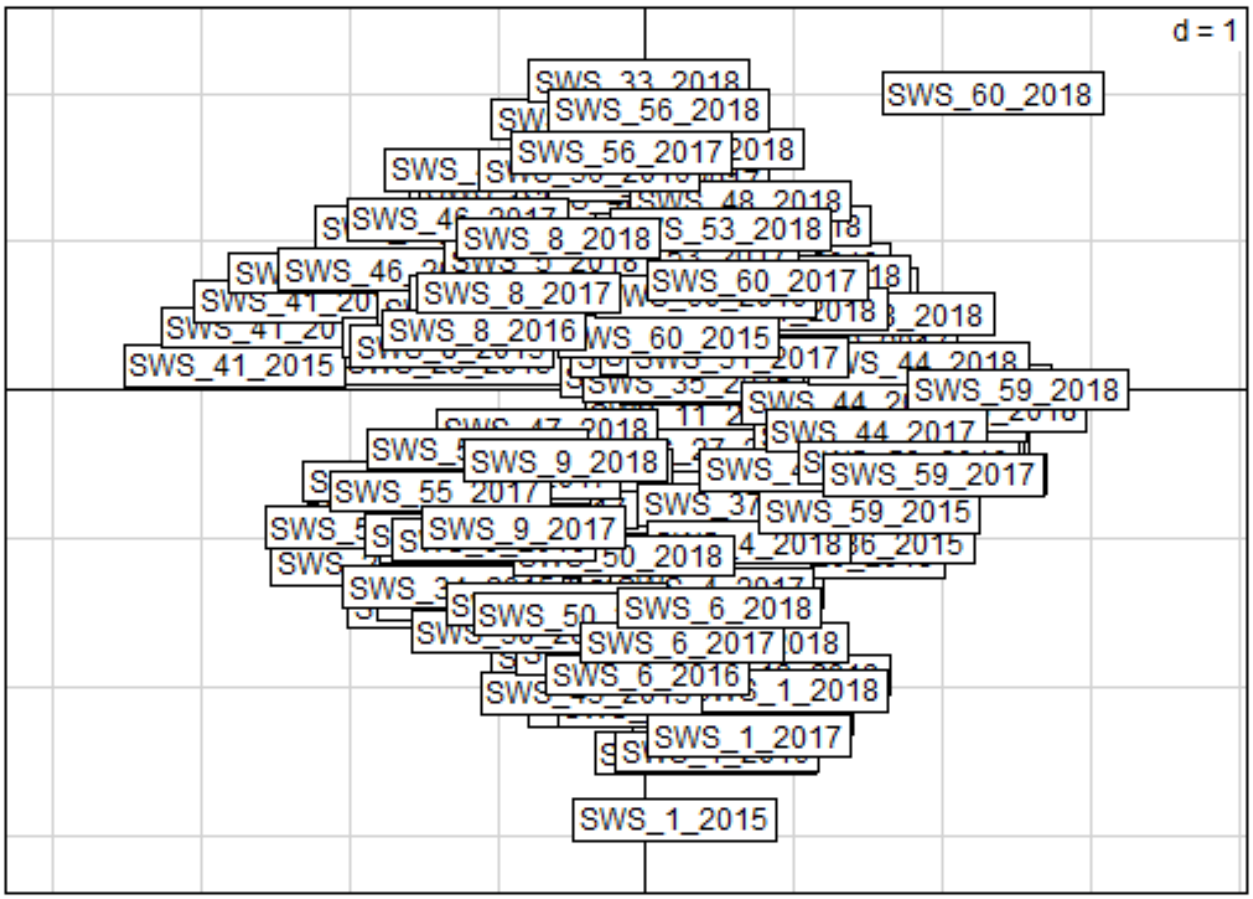
```
##                            sla       height     seedmass
## year                 0.09322166 -0.059905982  0.049384841
## treat.fall_12       -0.42830657  0.357916012 -0.347077743
## treat.fall_18       -0.12918333  0.324252965  0.372959091
## treat.fall_33        0.56170091  0.262035757  0.253846180
## treat.fall_6        -0.05107228  0.147035407  0.272959310
## treat.spring_12     -0.27963458 -0.199712799 -0.346269087
## treat.spring_18      0.23905949 -0.085845806 -0.008955102
## treat.spring_33      0.21167335 -0.144165313  0.039626404
## treat.spring_6      -0.48762201 -0.228927549  0.133997412
## water_cap           -0.03889957  0.001478144 -0.173512078
## cover_non_seeded     0.04607073  0.038685842 -0.049621504
## cover_seeded_grass   0.07007570 -0.013209380  0.019660490
```

RLQ Plots

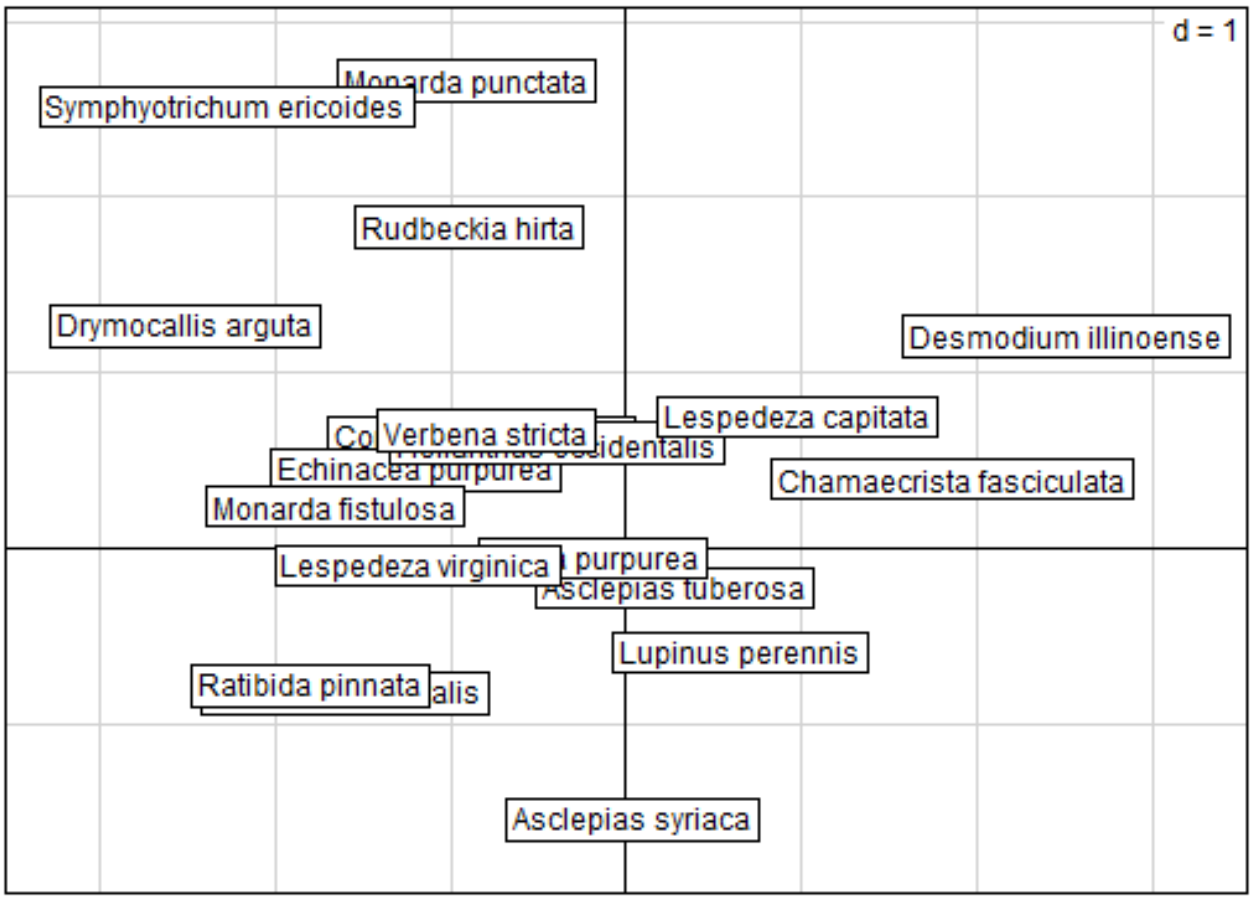
```r
plot(m_rlq)
```



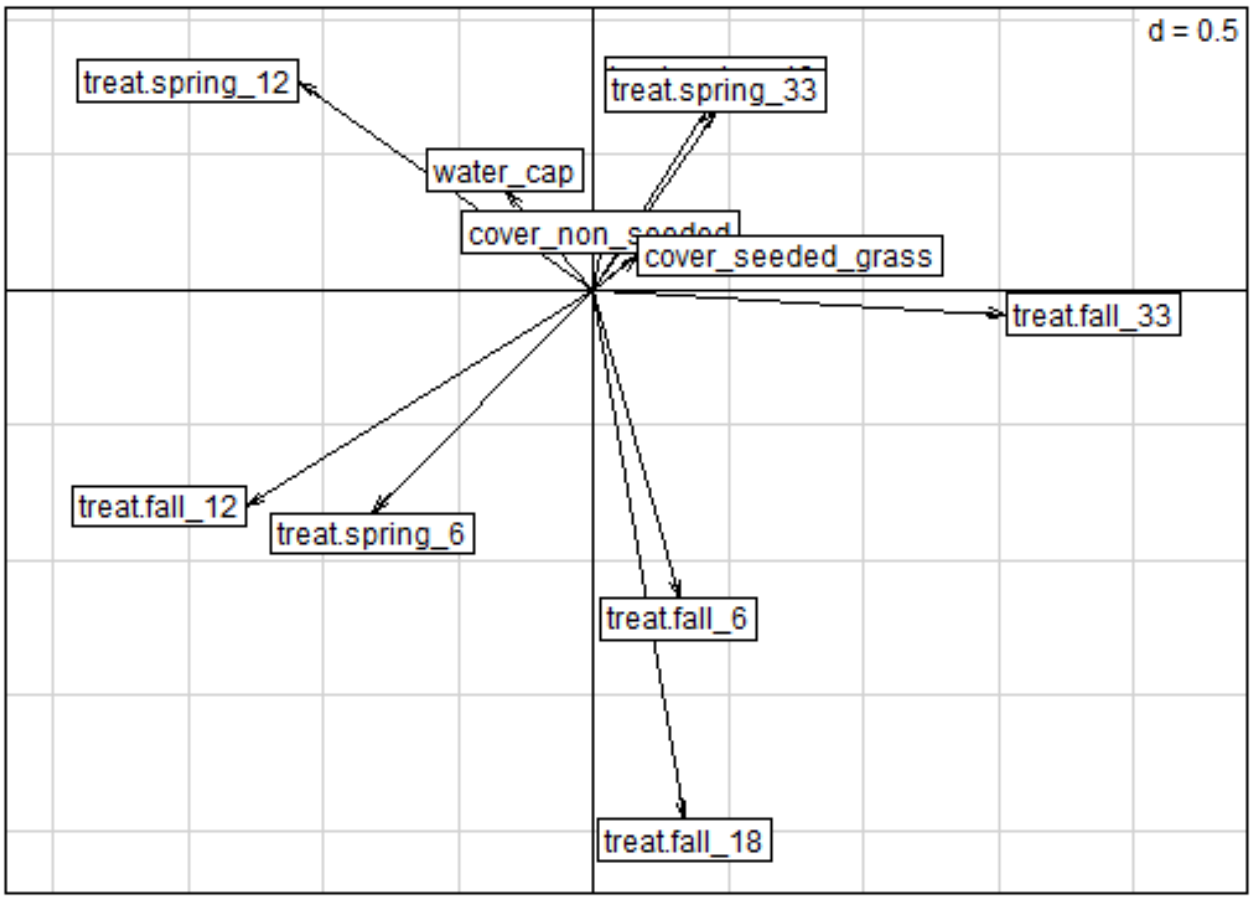

```
s.label(m_rlq$lR)
```



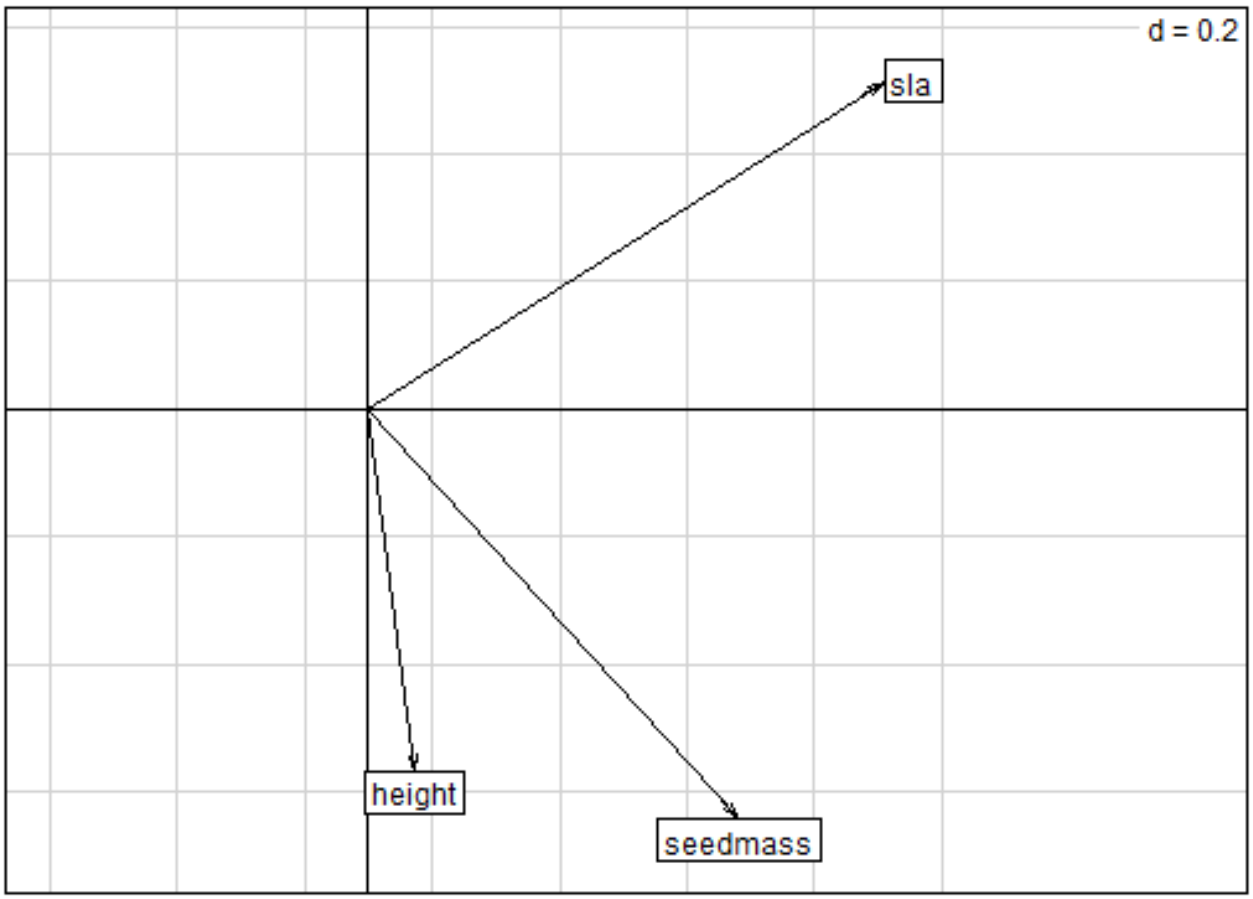

```
s.label(m_rlq$lQ)
```




```
s.arrow(m_rlq$l1)
```




```
s.arrow(m_rlq$c1)
```



Fourth-corner analysis

```
m_4th <- fourthcorner(
  tabR = sites,
  tabL = species,
  tabQ = traits,
  modeltype = 6,
  p.adjust.method.G = "none",
  p.adjust.method.D = "none",
  nrepet = 49999
)
```

```
m_4th_adj <- p.adjust.4thcorner(
  m_4th,
  p.adjust.method.G = "fdr",
  p.adjust.method.D = "fdr",
  p.adjust.D = "global"
)
m_4th_adj
```

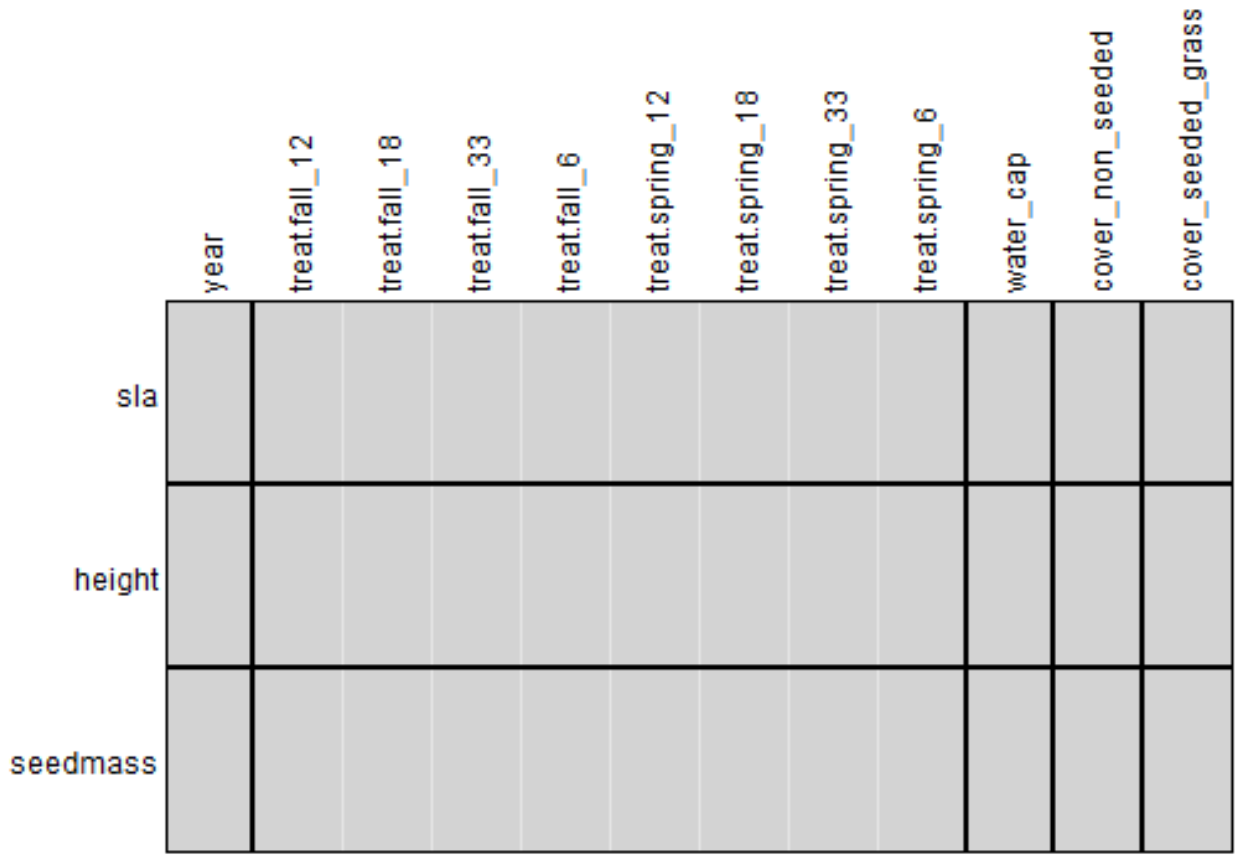
```
## Fourth-corner Statistics
## ------------------------
## Permutation method  Comb. 2 and 4  ( 49999  permutations)
##
## Adjustment method for multiple comparisons:   fdr
```

```
## call:  fourthcorner(tabR = sites, tabL = species, tabQ = traits, modeltype = 6,      nrepet = 49999,
##
## ---
##
##                              Test   Stat          Obs       Std.Obs     Alter
## 1                     year / sla      r  0.093221657  1.688408e+00 two-sided
## 2           treat.fall_12 / sla Homog.  0.042040616 -1.216081e+00      less
## 3           treat.fall_18 / sla Homog.  0.041654078 -7.451417e-01      less
## 4           treat.fall_33 / sla Homog.  0.252545522  4.643418e+00      less
## 5            treat.fall_6 / sla Homog.  0.044633336  8.721835e-02      less
## 6          treat.spring_12 / sla Homog.  0.051384326 -1.736298e+00      less
## 7          treat.spring_18 / sla Homog.  0.119402491 -4.230355e-01      less
## 8          treat.spring_33 / sla Homog.  0.304724022  6.201404e+00      less
## 9           treat.spring_6 / sla Homog.  0.032637759 -9.002019e-01      less
## 10               water_cap / sla      r -0.038899568 -6.229997e-01 two-sided
## 11        cover_non_seeded / sla      r  0.046070734  9.321029e-01 two-sided
## 12      cover_seeded_grass / sla      r  0.070075703  1.970790e+00 two-sided
## 13                 year / height      r -0.059905982 -1.088887e+00 two-sided
## 14        treat.fall_12 / height Homog.  0.129490818  1.141979e+00      less
## 15        treat.fall_18 / height Homog.  0.089897896  9.545535e-01      less
## 16        treat.fall_33 / height Homog.  0.108120603  5.290626e-02      less
## 17         treat.fall_6 / height Homog.  0.062940459  1.013431e+00      less
## 18      treat.spring_12 / height Homog.  0.162948265  1.513680e+00      less
## 19      treat.spring_18 / height Homog.  0.106983717 -1.078922e+00      less
## 20      treat.spring_33 / height Homog.  0.182054598  3.066557e+00      less
## 21       treat.spring_6 / height Homog.  0.109115445  9.251636e-01      less
## 22            water_cap / height      r  0.001478144  9.892621e-05 two-sided
## 23     cover_non_seeded / height      r  0.038685842  8.045717e-01 two-sided
## 24   cover_seeded_grass / height      r -0.013209380 -4.830802e-01 two-sided
## 25               year / seedmass      r  0.049384841  8.923313e-01 two-sided
## 26      treat.fall_12 / seedmass Homog.  0.142754121  1.348657e+00      less
## 27      treat.fall_18 / seedmass Homog.  0.047087640 -7.863496e-01      less
## 28      treat.fall_33 / seedmass Homog.  0.097867850 -2.130605e-01      less
## 29       treat.fall_6 / seedmass Homog.  0.062106815  9.039846e-01      less
## 30    treat.spring_12 / seedmass Homog.  0.163761683  9.027095e-01      less
## 31    treat.spring_18 / seedmass Homog.  0.165145056  1.182499e+00      less
## 32    treat.spring_33 / seedmass Homog.  0.190006282  2.042928e+00      less
## 33     treat.spring_6 / seedmass Homog.  0.073041025 -1.165797e-01      less
## 34          water_cap / seedmass      r -0.173512078 -2.522179e+00 two-sided
## 35   cover_non_seeded / seedmass      r -0.049621504 -1.047215e+00 two-sided
## 36 cover_seeded_grass / seedmass      r  0.019660490  4.308530e-01 two-sided
##     Pvalue Pvalue.adj
## 1  0.09652   0.694944
## 2  0.06502    0.58518
## 3    0.256    0.99396
## 4        1          1
## 5   0.5646          1
## 6  0.02784    0.47592
## 7  0.36466    0.99396
## 8        1          1
## 9  0.22478    0.99396
## 10 0.54212          1
## 11 0.37502    0.99396
## 12 0.03966    0.47592
```

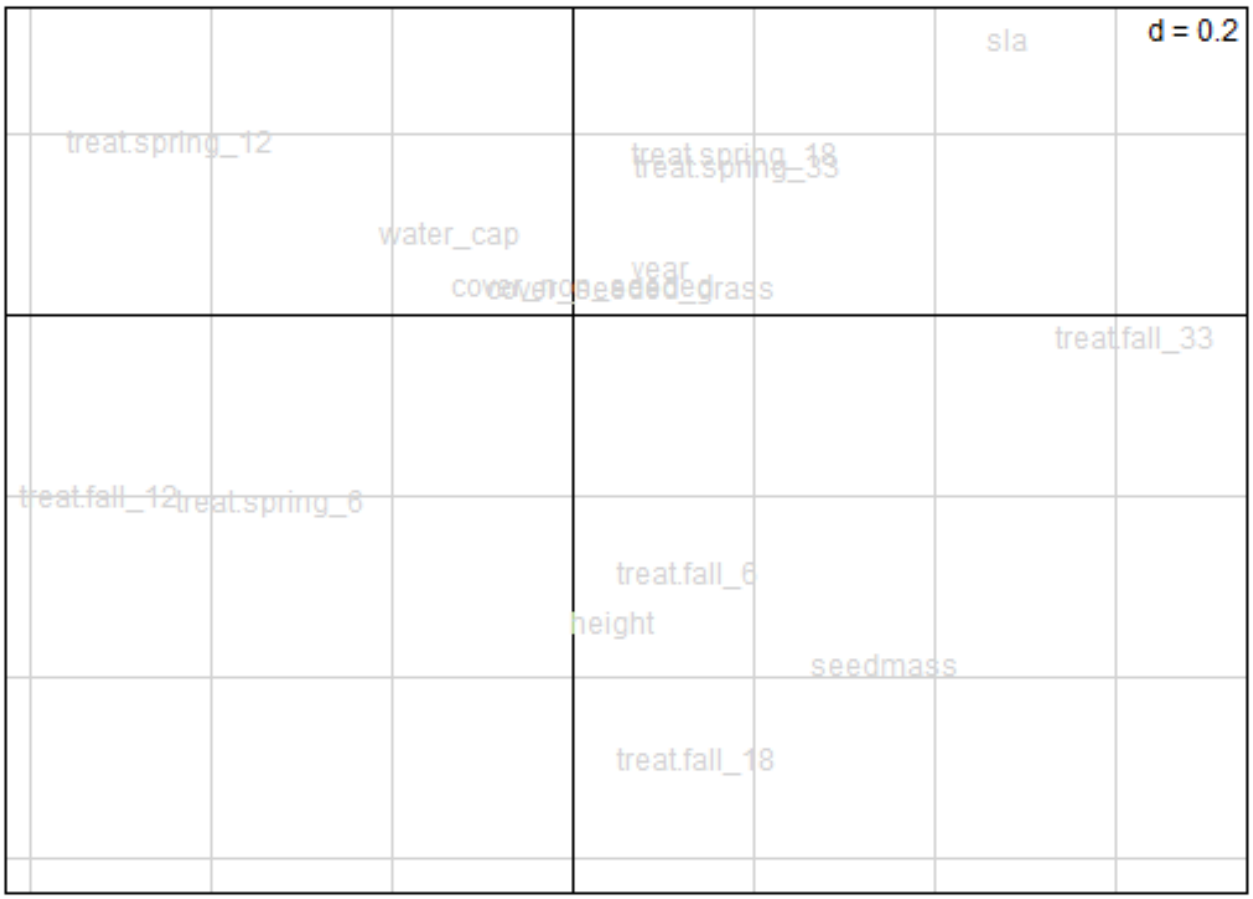
```
## 13 0.29474    0.99396
## 14 0.88168          1
## 15 0.82572          1
## 16 0.58804          1
## 17 0.82186          1
## 18 0.93048          1
## 19 0.14356    0.86136
## 20 0.99842          1
## 21  0.8155          1
## 22  0.9998          1
## 23  0.4698          1
## 24 0.62988          1
## 25 0.38654    0.99396
## 26 0.90352          1
## 27 0.23416    0.99396
## 28  0.4937          1
## 29 0.80112          1
## 30 0.81398          1
## 31 0.87568          1
## 32 0.97408          1
## 33 0.49592          1
## 34 0.00984    0.35424
## 35 0.32536    0.99396
## 36 0.67372          1
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Plot

```r
plot(m_4th_adj, alpha = 0.05, stat = "D2")
```




```r
plot(m_4th_adj,
  x.rlq = m_rlq,
  alpha = 0.05,
  stat = "D2",
  type = "biplot"
)
```



Save

Session info

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] ade4_1.7-23     lubridate_1.9.4 forcats_1.0.0   stringr_1.5.1
```

```
##  [5] dplyr_1.1.4     purrr_1.1.0     readr_2.1.5     tidyr_1.3.1
##  [9] tibble_3.3.0    ggplot2_3.5.2   tidyverse_2.0.0 here_1.0.1
## [13] tinytex_0.57
##
## loaded via a namespace (and not attached):
##  [1] bit_4.6.0          gtable_0.3.6       crayon_1.5.3       compiler_4.5.1
##  [5] Rcpp_1.1.0         tidyselect_1.2.1   parallel_4.5.1     scales_1.4.0
##  [9] yaml_2.3.10        fastmap_1.2.0      R6_2.6.1           generics_0.1.4
## [13] knitr_1.50         MASS_7.3-65        rprojroot_2.1.1    pillar_1.11.0
## [17] RColorBrewer_1.1-3 tzdb_0.5.0         rlang_1.1.6        utf8_1.2.6
## [21] stringi_1.8.7      xfun_0.52          bit64_4.6.0-1      timechange_0.3.0
## [25] cli_3.6.5          withr_3.0.2        magrittr_2.0.3     digest_0.6.37
## [29] grid_4.5.1         vroom_1.6.5        rstudioapi_0.17.1  hms_1.1.3
## [33] lifecycle_1.0.4    vctrs_0.6.5        evaluate_1.0.4     glue_1.8.0
## [37] farver_2.1.2       rmarkdown_2.29     tools_4.5.1        pkgconfig_2.0.3
## [41] htmltools_0.5.8.1
```